

HELIX AI

Agentic AI Healthcare Companion

IBM SkillsBuild Project

Project Domain	Artificial Intelligence in Healthcare
Core Technologies	IBM watsonx Orchestrate, IBM Granite, RAG, Flask
Project Focus	Chronic Health Monitoring and Personalized Health Guidance

1. Problem Statement

Chronic disease management requires continuous observation of health indicators, timely recognition of abnormal changes, consistent treatment adherence, and personalized lifestyle support. In conventional healthcare workflows, patients often receive guidance only during scheduled consultations. Between visits, they may struggle to interpret changing vital signs, understand treatment information, or identify when symptoms require professional attention.

2. Identified Challenge

Existing general-purpose digital assistants commonly provide broad health information without considering a user's recorded vitals, historical health profile, medication context, or risk trends. Health data is also frequently fragmented across separate records and monitoring systems. This creates a need for a context-aware healthcare support platform capable of coordinating specialized tasks while maintaining safe, non-diagnostic guidance.

3. Proposed Solution

Helix AI is an agentic AI healthcare companion designed to provide continuous, personalized, and context-aware health support. IBM watsonx Orchestrate coordinates a Supervisor Agent and specialized Clinical Assessment, Chronic Monitoring, Lifestyle, Treatment, and Health Report agents. Each request is routed to the appropriate specialist based on user intent.

4. Technical Approach

The platform combines IBM Granite foundation models through watsonx.ai, secure OpenAPI-based patient-data tools, deterministic risk-analysis logic, and a Retrieval-Augmented Generation (RAG) medical knowledge base. This hybrid architecture allows Helix AI to use patient context and grounded medical information while preserving explicit safety rules and non-diagnostic behavior.

5. Project Objectives

- Provide personalized health guidance using recorded patient profile and vital-sign context.
- Coordinate specialized healthcare tasks through a multi-agent architecture.
- Identify abnormal vital patterns using deterministic risk classification and red-flag logic.
- Ground educational health information using curated medical documents and RAG retrieval.

- Support lifestyle management, medication awareness, and treatment adherence tracking.
- Generate concise health report snapshots by combining profile, vitals, risk, lifestyle, and timeline data.

6. Scope

The current implementation focuses on chronic health monitoring for conditions such as diabetes, hypertension, and heart disease. The medical knowledge base additionally supports broader educational health queries and symptom-related information. Helix AI is intended to assist users in understanding their recorded health information and does not replace clinical diagnosis or professional medical decision-making.

7. Expected Impact

Helix AI demonstrates how agentic AI can transform a single conversational interface into a coordinated healthcare support system. By integrating multi-agent orchestration, grounded medical retrieval, patient-context tools, and deterministic safety logic, the project aims to improve health awareness, encourage consistent monitoring, and provide clearer, more relevant educational guidance between healthcare interactions.

Medical Disclaimer: Helix AI is an educational IBM SkillsBuild project. It is not a medical device, does not provide definitive diagnoses, and must not replace qualified healthcare professionals or emergency services.